

STAT 112 : Introduction to Statistics and Probability II

LECTURE 1

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Definition 1.1 Discrete Random Variable

A random variable X is said to be discrete if it can assume only finite or countably infinite number of distinct values.

Notationally, we use uppercase letters, such as X , Y or Z to denote a random variable and we use lowercase letters such as x , y or z to denote a particular value.

Definition 1.2 Random Variable

A random variable (r.v) is a real-valued function with a sample space as domain: $X : S \mapsto \mathbb{R}$ where the set of real numbers $\mathbb{R} = (-\infty, \infty)$.

Definition 1.3 Discrete Probability Function

Let S be the sample space and let X be a random variable. Then the probability function or (probability mass function) for the r.v X is given by

$$P_x(X = x_i) \equiv P(X = x_i) = P_s(\{s \in S : X(s) = x_i\})$$

The sample space of X is

$$S_x = \{x_i \in \mathbb{R} : \text{there exists an } s \in S \text{ with } X(s) = x_i\}$$

The expression $(X = x_i)$ can be read as the set of all points in S assigned the value x_i by the random variable X .

Example 1.1

A supermarket has three men and three women working as salespersons. Suppose two salespersons are to be selected for a special job. Let X denote the number of women selected. Find the probability distribution for X .

Solution

There are $\binom{6}{2} = 15$ ways of selecting two salespersons among the six salespersons. Here S contains 15 sample points. The values for X are 0, 1 and 2.

The number of ways of selecting $X = 0$ women is $\binom{3}{0} \binom{3}{2}$

The number of ways of selecting $X = 1$ women is $\binom{3}{1} \binom{3}{1}$

The number of ways of selecting $X = 2$ women is $\binom{3}{2} \binom{3}{0}$

Hence

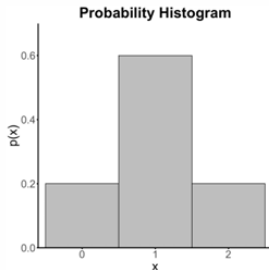
$$P(0) = P(X = 0) = \frac{\binom{3}{0} \binom{3}{2}}{\binom{6}{2}} = \frac{3}{15} = \frac{1}{5}$$

$$P(1) = P(X = 1) = \frac{\binom{3}{1} \binom{3}{1}}{\binom{6}{2}} = \frac{9}{15} = \frac{3}{5}$$

$$P(2) = P(X = 2) = \frac{\binom{3}{2} \binom{3}{0}}{\binom{6}{2}} = \frac{3}{15} = \frac{1}{5}$$

The probability distribution for Example 1.1 is given as:

X	0	1	2
$P_X(x)$	$\frac{1}{5}$	$\frac{3}{5}$	$\frac{1}{5}$



The area under the bar is equal to the probability that X takes on the indicated value over which the bar is centered.

The most concise method of representing discrete probability distributions is by means of a formula. For the problem in Example 1.1, we can represent

$$P(x) = \frac{\binom{3}{x} \binom{3}{2-x}}{\binom{6}{2}}, \quad x = 0, 1, 2$$

Theorem 1.1

For any discrete probability distribution, the following must be true:

- (i) $0 \leq P(x) \leq 1$ for all x
- (ii) $\sum_{\text{all } x} P(x) = 1$, where the summation is over all values of x with non-zero probability.

Definition 1.4

Let X be a discrete random variable with the probability function $p(x)$.

Then the expected value of X , denoted by $E(X)$, is defined as

$$E(X) = \sum_{\text{all } x} xp(x)$$

Theorem 1.2

Let X be a discrete random variable with probability function $p(x)$ and $g(X)$ be a real-valued function of X . Then the expected value of $g(X)$ is given by

$$E[g(X)] = \sum_{\text{all } x} g(x)p(x)$$

Definition 1.5

If X is a random variable with mean $E(X) = \mu$, the variance of a random variable X is defined to be the expected value of $(x - \mu)^2$. That is,

$$\text{Var}(X) = E[(X - \mu)^2]$$

The standard deviation of X is the positive square root of $\text{Var}(X)$.

$$\text{SD}(X) = +\sqrt{\text{Var}(X)}$$

Example 1.2

Given that the r.v X has the probability mass function with the following data as given in the table:

x	0	1	2	3
$P_X(x)$	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{3}{8}$	$\frac{1}{4}$

Find the mean, variance and standard deviation of X .

Solution

By definition of the mean and variance,

$$\mu = E(X) = \sum_{x=0}^3 xp(x) = 0 \left(\frac{1}{8} \right) + 1 \left(\frac{1}{4} \right) + 2 \left(\frac{3}{8} \right) + 3 \left(\frac{1}{4} \right)$$

$$\mu = 1.75$$

$$\begin{aligned}
 \sigma^2 = \text{Var}(X) &= E[(X - \mu)^2] = \sum_{x=0}^3 (x - 1.75)^2 p(x) \\
 &= (0 - 1.75)^2 \left(\frac{1}{8}\right) + (1 - 1.75)^2 \left(\frac{1}{4}\right) \\
 &\quad + (2 - 1.75)^2 \left(\frac{3}{8}\right) + (3 - 1.75)^2 \left(\frac{1}{4}\right)
 \end{aligned}$$

$$\sigma^2 = 0.9375$$

$$SD(X) = +\sqrt{\text{Var}(X)} = \sqrt{0.9375} = 0.97$$

Theorem 1.3

Let X be a discrete r.v with pmf $p(x)$ and c be a constant. Then $E(c) = c$

Proof

Consider the function $g(X) = c$, then by Theorem 1.2, we have

$$E(c) = \sum_{\text{all } x} cp(x) = c \sum_{\text{all } x} p(x)$$

But $\sum_{\text{all } x} p(x) = 1$, and hence $E(c) = c$

Theorem 1.4

Let X be a discrete random variable with pmf $p(x)$, $g(X)$ be any function of X , and c be a constant. Then

$$E[cg(X)] = cE[g(X)]$$

Proof

By Theorem 1.2,

$$E[cg(X)] = \sum_{\text{all } x} cg(x)p(x) = c \sum_{\text{all } x} g(x)p(x) = cE[g(X)]$$

Theorem 1.5

Let X be a discrete r.v with pmf $p(x)$ and $g_1(X), g_2(X), \dots, g_k(X)$ be k functions of X . Then

$$E[g_1(X) + g_2(X) + \dots + g_k(X)] = E[g_1(X)] + E[g_2(X)] + \dots + E[g_k(X)]$$

$$\text{ie. } E \left[\sum_{i=1}^k g_i(X) \right] = \sum_{i=1}^k E[g_i(X)]$$

Theorem 1.6

Let X be a discrete r.v with pmf $p(x)$ and mean $E(X) = \mu$: then

$$\text{Var}(X) = \sigma^2 = E[(X - \mu)^2] = E(X^2) - \mu^2$$

Proof

$$\begin{aligned}\sigma^2 &= E[(X - \mu)^2] = E[X^2 - 2\mu X + \mu^2] \\&= E(X^2) - 2\mu E(X) + \mu^2 \\&= E(X^2) - 2\mu^2 + \mu^2 \\&= E(X^2) - \mu^2\end{aligned}$$

Example 1.4

Let X be a discrete r.v with pmf $p(x)$ given the accompanying table:

x	1	2	3	4
$p(x)$	0.4	0.3	0.2	0.1

Find $E(X)$, $E\left(\frac{1}{X}\right)$, $E(X^2 - 1)$ and $Var(X)$.

Solution

By definition

$$\begin{aligned} E(X) &= \sum_{\text{all } x} xp(x) = 1(0.4) + 2(0.3) + 3(0.2) + 4(0.1) \\ &= 0.4 + 0.6 + 0.6 + 0.4 = 2 \end{aligned}$$

$$E\left(\frac{1}{X}\right) = \sum_{\text{all } x} \frac{1}{x} p(x)$$

$$= 1(0.4) + \frac{1}{2}(0.3) + \frac{1}{3}(0.2) + \frac{1}{4}(0.1)$$

$$E\left(\frac{1}{X}\right) = 1$$

$$E(X^2 - 1) = \sum_{\text{all } x} (x^2 - 1)p(x)$$

$$= (1^2 - 1)(0.4) + (2^2 - 1)(0.3) + (3^2 - 1)(0.2) + (4^2 - 1)(0.1)$$

$$E(X^2 - 1) = 4$$

$$\text{Var}(X) = E(X^2) - E^2(X) = E(X^2) - 2^2$$

But

$$E(X^2) = \sum_{\text{all } x} x^2 p(x)$$

$$= 1^2(0.4) + 2^2(0.3) + 3^2(0.2) + 4^2(0.1) = 5$$

$$\therefore \text{Var}(X) = 5 - 4 = 1$$